Advanced Programming

Lab 12, class inheritance & polymorphism

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Exercise:

1. Please find the errors of the following code and explain why to SA.

```
class Base
{
   private:
      int x;
   protected:
      int y;
   public:
      int z;
   void funBase (Base& b)
   {
      ++x;
      ++y;
      ++z;
      ++b.x;
      ++b.y;
      ++b.z;
    }
};
```

```
class Derived:public Base
{
   public:
      void funDerived (Base& b, Derived& d)
      {
         ++x;
         ++y;
         ++z;
         ++b.x;
         ++b.y;
         ++b.z;
         ++d.x;
         ++d.y;
         ++d.z;
      }
};
```

```
void fun(Base& b, Derived& d)
{
   ++x;
   ++y;
   ++z;
   ++b.x;
   ++b.y;
   ++b.z;
   ++d.x;
   ++d.y;
   ++d.z;
}
```

Exercise:

2. Run the following program, and explain the result to SA.

```
#include <iostream>
using namespace std;

class Polygon
{
  protected:
    int width, height;
  public:
    void set_values (int a, int b)
      { width=a; height=b; }
    int area()
    { return 0;}
};
```

```
class Rectangle: public Polygon
{
  public:
    int area()
      { return width * height; }
};

class Triangle: public Polygon
{
  public:
    int area()
      { return width*height/2; }
};
```

```
int main ()
{
  Rectangle rect;
  Triangle trgl;
  Polygon * ppoly1 = &rect;
  Polygon * ppoly2 = &trgl;
  ppoly1->set_values (4,5);
  ppoly2->set_values (2,5);

  cout << rect.area() << endl;
  cout << trgl.area() << endl;
  cout << ppoly1->area() << endl;
  cout << ppoly2->area() << endl;
  return 0;
}
```

Exercise:

3. Run the following program, and explain the result to SA. Is there any problem in the program?

```
// dynamic allocation and polymorphism
#include <iostream>
using namespace std;

class Polygon
{
  protected:
    int width, height;
  public:
    Polygon (int a, int b) : width(a), height(b) {}
    virtual int area (void) =0;
    void printarea()
      { cout << this->area() << '\n'; }
};
```

```
class Rectangle: public Polygon
{
  public:
    Rectangle(int a,int b) : Polygon(a,b) {}
    int area()
      { return width*height; }
};

class Triangle: public Polygon
{
  public:
    Triangle(int a,int b) : Polygon(a,b) {}
    int area()
      { return width*height/2; }
};
```

```
int main ()
{
  Polygon * ppoly = new Rectangle (4,5);
  ppoly->printarea();
  ppoly = new Triangle (2,5);
  ppoly->printarea();

  return 0;
}
```